

Sums

Outline

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Definition

Introduction: Definition

- ▶ In this section we introduce the **sum symbol** $\sum$, a compact notation for the sum of many terms
- ▶ It is very useful when the terms are given by a formula depending on an index
- ▶ Sums appear everywhere in linear algebra: scalar products, matrix products, norms, ...

Definition 1: sum

Given n values $a_j \in \mathbb{R}$ with $j \in \{1, 2, \dots, n\}$, the **sum**

$$a_1 + a_2 + \dots + a_n$$

can be indicated in compact form with the sum symbol:

$$\sum_{j=1}^n a_j$$

- ▶ The symbol j is called **sum index**.
- ▶ The sum $\sum_{j=1}^n a_j$ reads: “sum for j from 1 to n of a_j ”.
- ▶ The sum symbol is very useful when the terms a_j are explicitly defined as a function of the sum index j .

Example

$$\sum_{j=1}^{10} \frac{1}{j} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{j=3}^n j^2 = 3^2 + 4^2 + 5^2 + \dots + n^2$$

- ▶ The sum index is a **dummy index**. This means that if j is replaced with i , k or any other index in all its occurrences the value of the sum does not change.

Example: dummy index

- We have:

$$\sum_{j=1}^n j^2 = \sum_{i=1}^n i^2 = \sum_{k=1}^n k^2$$

as the symbols indicate the sum of the squares of the first n natural numbers (without zero).

- On the contrary, we have:

$$\sum_{j=1}^n j^2 \neq \sum_{j=1}^m j^2$$

as the two symbols indicate the sum, respectively, of the first n and the first m squares of the natural numbers (without zero). Clearly, if $n \neq m$, the result is different.

Summary: Definition

Sum symbol:

- ▶ $\sum_{j=1}^n a_j$ is a compact notation for $a_1 + a_2 + \cdots + a_n$
- ▶ j is the sum index; 1 and n are its first and last value

Dummy index:

- ▶ Renaming the index (e.g. $j \rightarrow i$) does not change the value of the sum
- ▶ Changing the limits (e.g. $n \rightarrow m$) does change it

Properties

Introduction: Properties

- ▶ Now we will explore six fundamental properties of sums
- ▶ They will help us manipulate and simplify summation expressions
- ▶ Three of them are algebraic (constant terms, constant factors, union of sums), three concern the index (decomposition, translation, reflexion)

Observation 1: sum with constant term

For every natural number $n \geq 1$ and real number c , we have:

$$\sum_{j=1}^n c = n c \quad (1)$$

Proof

We have:

$$\sum_{j=1}^n c = \underbrace{c + c + \dots + c}_{c \text{ summed up } n \text{ times}}$$
$$= n c$$



Observation 2: product by a constant

For every sum $\sum_{j=1}^n a_j$ and real number c , we have:

$$\sum_{j=1}^n (c a_j) = c \sum_{j=1}^n a_j \quad (2)$$

Proof

From the distributive property, we have:

$$\sum_{j=1}^n (c a_j) = c a_1 + c a_2 + \dots + c a_n$$

$$= c (a_1 + a_2 + \dots + a_n)$$

$$= c \sum_{j=1}^n a_j$$



Observation 3: union of sums

For every pair of sums $\sum_{j=1}^n a_j$ and $\sum_{j=1}^n b_j$, we have:

$$\sum_{j=1}^n a_j + \sum_{j=1}^n b_j = \sum_{j=1}^n (a_j + b_j) \quad (3)$$

Proof

We have:

$$\begin{aligned}\sum_{j=1}^n a_j + \sum_{j=1}^n b_j &= a_1 + a_2 + \dots + a_n + b_1 + b_2 + \dots + b_n \\ &= a_1 + b_1 + a_2 + b_2 + \dots + a_n + b_n \\ &= \sum_{j=1}^n (a_j + b_j)\end{aligned}$$



Observation 4: decomposition

For every sum $\sum_{j=1}^{n+m} a_j$, we have:

$$\sum_{j=1}^{n+m} a_j = \sum_{j=1}^n a_j + \sum_{j=n+1}^{n+m} a_j \quad (4)$$

Observation 5: index translation

For every sum $\sum_{j=1}^n a_j$ and natural number $m \geq 1$, we have:

$$\sum_{j=1}^n a_j = \sum_{j=1+m}^{n+m} a_{j-m} = \sum_{j=1-m}^{n-m} a_{j+m} \quad (5)$$

Observation 6: index reflexion

For every sum $\sum_{j=1}^n a_j$, we have:

$$\sum_{j=1}^n a_j = \sum_{j=1}^n a_{n-j+1} = \sum_{j=0}^{n-1} a_{n-j} \quad (6)$$

Proof

The three properties (decomposition, index translation and index reflexion) are simply different notations and/or arrangements of the terms in the sums.



Summary: Properties (1/2)

Algebraic properties:

- ▶ Constant term: $\sum_{j=1}^n c = n c$
- ▶ Product by a constant: $\sum_{j=1}^n (c a_j) = c \sum_{j=1}^n a_j$
- ▶ Union of sums: $\sum_{j=1}^n a_j + \sum_{j=1}^n b_j = \sum_{j=1}^n (a_j + b_j)$

Summary: Properties (2/2)

Index properties:

- ▶ Decomposition: $\sum_{j=1}^{n+m} a_j = \sum_{j=1}^n a_j + \sum_{j=n+1}^{n+m} a_j$
- ▶ Index translation: $\sum_{j=1}^n a_j = \sum_{j=1+m}^{n+m} a_{j-m}$
- ▶ Index reflexion: $\sum_{j=1}^n a_j = \sum_{j=1}^n a_{n-j+1}$

- ▶ They only rearrange or rename the terms of the sum

Some Important Sums



Introduction: Some important sums

- ▶ We now present three classical formulas for important sums that appear frequently in mathematics and computer science
- ▶ The sum of the first n positive natural numbers, of the first n positive even numbers and of the first n odd numbers
- ▶ The proofs are a good exercise in using the properties of the previous section

Observation 7: sum of the first n positive natural numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=1}^n j = \frac{n^2 + n}{2} \quad (7)$$

Proof (1/2)

Using index reflexion in the second sum, we have:

$$\begin{aligned}\sum_{j=1}^n j &= \frac{1}{2} \left(\sum_{j=1}^n j + \sum_{j=1}^n j \right) \\ &= \frac{1}{2} \left(\sum_{j=1}^n j + \sum_{j=1}^n (n - j + 1) \right) \\ &= \frac{1}{2} \sum_{j=1}^n (j + n - j + 1)\end{aligned}$$

Proof (2/2)

Continuing:

$$\begin{aligned}\sum_{j=1}^n j &= \frac{1}{2} \sum_{j=1}^n (j + n - j + 1) \\ &= \frac{1}{2} \sum_{j=1}^n (n + 1) \\ &= \frac{n(n + 1)}{2} \\ &= \frac{n^2 + n}{2}\end{aligned}$$



Observation 8: sum of the first n positive even numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=1}^n 2j = n^2 + n \quad (8)$$

Proof

We have:

$$\begin{aligned}\sum_{j=1}^n 2j &= 2 \sum_{j=1}^n j \\ &= 2 \left(\frac{n^2 + n}{2} \right) \\ &= n^2 + n\end{aligned}$$



Observation 9: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{j=0}^{n-1} (2j+1) = n^2$$

or equivalently

$$\sum_{j=1}^n (2j-1) = n^2$$

Proof (1/3)

We have:

$$\begin{aligned}\sum_{j=0}^{n-1} (2j + 1) &= 2 \sum_{j=0}^{n-1} j + \sum_{j=0}^{n-1} 1 \\ &= 2 \sum_{j=1}^n (j - 1) + \sum_{j=1}^n 1 \\ &= 2 \left(\sum_{j=1}^n j - \sum_{j=1}^n 1 \right) + n\end{aligned}$$

Proof (2/3)

Continuing:

$$\begin{aligned}\sum_{j=0}^{n-1} (2j+1) &= 2 \left(\sum_{j=1}^n j - \sum_{j=1}^n 1 \right) + n \\ &= 2 \left(\frac{n^2+n}{2} - n \right) + n \\ &= n^2 + n - 2n + n \\ &= n^2\end{aligned}$$

Proof (3/3)

Equivalently:

$$\begin{aligned}\sum_{j=1}^n (2j - 1) &= 2 \sum_{j=1}^n j - \sum_{j=1}^n 1 \\ &= 2 \left(\frac{n^2 + n}{2} \right) - n \\ &= n^2 + n - n \\ &= n^2\end{aligned}$$



Summary: Some important sums

Classical formulas (for every $n \geq 1$):

- ▶ $\sum_{j=1}^n j = \frac{n^2+n}{2}$ (first n positive natural numbers)
- ▶ $\sum_{j=1}^n 2j = n^2 + n$ (first n positive even numbers)
- ▶ $\sum_{j=1}^n (2j-1) = n^2$ (first n odd numbers)

Proof technique:

- ▶ Write the sum twice, reflect the index in one copy, and add term by term

Key Takeaways

Summary

1. Notation:

- ▶ The sum symbol $\sum_{j=1}^n a_j$ provides compact notation for $a_1 + a_2 + \cdots + a_n$
- ▶ The index is a dummy variable

2. Key Properties:

- ▶ Linearity: $\sum (a_j + b_j) = \sum a_j + \sum b_j$ and $\sum c \cdot a_j = c \cdot \sum a_j$
- ▶ Index manipulation: decomposition, translation and reflexion

3. Important Formulas:

- ▶ $\sum_{j=1}^n j = \frac{n^2+n}{2}, \quad \sum_{j=1}^n 2j = n^2 + n, \quad \sum_{j=1}^n (2j - 1) = n^2$